MOVIE RENTAL ANALYTICS

EDA • SQL Analysis • Power BI Desktop



Presentation roadmap

01 Business problem

Understand catalog, rentals, customers, stores and revenue.

02 MECE framework

Five analysis areas organize the work.

03 Data preparation

Validate types, keys, missing values and joins.

04 EDA & SQL

Explore patterns using PostgreSQL.

05 Power BI Desktop

Create KPIs, slicers and interactive visuals.

MECE framework

1. Movie analysis

Movie characteristics • Movie availability

2. Categories & content

Category performance • Content trends

3. Actors & customer

Actor performance • Customer behavior

4. Rental & revenue

Rental trends • Revenue performance

5. Store & global customer

Store performance • Geographical analysis

Dataset and table map

Area	Tables	Purpose / fields
Movies	film, film_text, language	Title, release year, rating, rental rate, length, language
Categories & actors	category, film_category, actor, film_actor	Genres and actor-to-film mapping
Rental activity	rentat, inventory	Rental date, customer, staff, inventory copy
Payments	payment	Amount, payment date, customer, staff, rental
Customers & geography	customer, address, city, country	Customer activity and location
Stores & staff	store, staff	Store address, staff assignment and rental operations

Data cleaning and validation

- Confirm table and column names using information_schema.
- Check key uniqueness: film_id, customer_id, inventory_id, rental_id, payment_id and mapping tables.
- Check missing values in dates, IDs, language, rental records and payment amounts.
- Convert text amounts before calculations: NULLIF(amount, '')::NUMERIC if values are valid numeric text.
- Convert text dates before date operations: rental_date::TIMESTAMP and create_date::TIMESTAMP when compatible.
- Check joins for duplicate rows; aggregate payments by rental first if multiple payment rows can exist.
- Limitations: no direct satisfaction score, customer age, staff knowledge score or geographic coordinates in the supplied schema.

Exploratory data analysis (EDA)

EDA area	Questions / checks
Movie catalog	How many films? Which ratings, release years, rental rates and lengths are common?
Availability	How many inventory copies per film? Which films have low or no inventory?
Category & actors	Which categories have largest collections? Which actors appear in the most films?
Customer behavior	How many active/inactive customers? How many one-time and repeat renters?
Rental & revenue	Which films are rented most? What are trends by month, weekday and hour?
Store & geography	Which stores, cities and countries contribute most rentals, customers and revenue?

SQL analysis workflow

1. Inspect

Verify schema, names, data types and row counts.

2. Clean

Cast text numbers/dates; check nulls and keys.

3. Join

Connect customer → rentat → inventory → film/category; payment → rental.

4. Aggregate

Calculate counts, distinct customers, revenue and time trends.

5. Validate

Compare totals with source tables and check duplicate joins.

6. Connect

Use PostgreSQL connector or export query results to CSV.

Core KPIs and definitions

KPI	Definition	Caution
Total films	Distinct film_id in film	Catalog size, not inventory copies
Inventory copies	Count of inventory_id	Listed copies, not live availability
Total rentals	Count of rental_id in rentat	Count unique rentals to avoid join duplication
Unique customers	Distinct customer_id with rental activity	Different from active status
Total revenue	Sum of numeric payment amount	Cast text to NUMERIC; check duplicate joins
Repeat customers	Customers with more than one rental	Define period for retention analysis
Inventory utilization	Inventory rented at least once ÷ total inventory	Not real-time availability

Power BI Desktop: load the data

- Open Power BI Desktop → Home → Get data.
- Choose PostgreSQL database and enter server/database details; or select Text/CSV for exported SQL results.
- Choose Import for a simple beginner workflow, unless your project requires DirectQuery.
- Select required tables and click Transform Data.
- In Power Query, check data types, trim text, replace blanks where appropriate and remove accidental duplicates.
- Click Close & Apply to load the model.
- Open Model view and create/verify relationships using matching ID fields.

Suggested relationships

From	Key / link	To	Purpose
film	film_id	inventory → film_id	Film represented by inventory copy
film	film_id	film_category → film_id	Films to categories
category	category_id	film_category → category_id	Category names
inventory	inventory_id	rentat → inventory_id	Rental activity to copy
customer	customer_id	rentat → customer_id	Customer rental history
rentat	rental_id	payment → rental_id	Rental-to-payment link
customer	address_id	address → address_id	Customer location
address	city_id	city → city_id → country_id	Geographical reporting
staff	staff_id	rentat → staff_id	Staff rental activity

Use one-to-many relationships where appropriate; check whether payment and rental joins multiply rows.

Power BI report page plan

Page	Recommended visuals	Purpose
1. Overview	Cards: films, inventory, rentals, customers, revenue; monthly trend	Business snapshot
2. Movies & categories	Top-rented film bar; films by category; rating/language slicers	Catalog and demand
3. Customers	New/repeat comparison; active status; rental frequency	Customer engagement
4. Rentals & revenue	Monthly revenue line; revenue by staff/store; day/hour matrix	Operations and revenue
5. Store & geography	City/country bars or map if valid geographic data is available	Location performance
6. Inventory	Inventory copies and rented-ever copies; low-use film detail table	Inventory utilization

Visual selection guide

Question	Visual	Example fields
What changes over time?	Line chart	Month on X; rentals/revenue on Y
Which films/categories rank highest?	Horizontal bar	Film/category; rental count
How do stores compare?	Clustered column	Store; revenue and rentals
What is the mix?	Donut or stacked bar	Category/customer type; count or revenue
When are rentals busiest?	Matrix heatmap	Day rows; hour columns; rental count
What needs investigation?	Table / matrix	Film, category, inventory, rentals, revenue

Sort bars descending, use descriptive titles and avoid donut charts with too many categories.

Interactivity and usability

- Add slicers for rental date/year, store, category, rating, language and customer active status where available.
- Use Edit interactions to confirm slicers filter the intended visuals.
- Add drill-through from category to film details or from store to store-level metrics.
- Format revenue as currency, counts as whole numbers and rates as percentages.
- Add tooltips with unique customers, inventory copies and rental count.
- Keep a clear layout: KPI cards at top, trends in the middle, detail tables below.
- Use meaningful titles and show the selected date range where possible.

THANK YOU.